ASD – AI

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What is AI?

Meaningless marketing term?

EU definition: anything you can do with computer

- Exact
  - Reasoning (Boolean logic)
  - Optimization (genetic algorithms, also reasoning)
- Fuzzy
  - Perception (Neural networks)
- Knowledge representation
  - Knowledge base
  - Ontologies

What is AI?

- Knowledge representation
  - Knowledge base
  - Ontologies
- Reasoning
  - Boolean logic (incl. defeasible reasoning)
  - Genetic algorithms
- Perception
  - Neural networks

Knowledge representation

Knowledge base *aka expert systems*

Ontologies:

Human curated representation of concepts
and the relations between them.

Strives to be exact, but
real-world concepts are inherently 'fuzzy'.

Does not scale, due to need to manual construction

Effectively dead, killed off by perception AI.

Reasoning

Boolean logic

Either exact: mathematically ideal model of real-world problem

Or defeasible: = exact + maybe.

Exact Boolean logic models a problem mathematically:
as a NP-hard or NP-complete representation, e.g.:

- Integer linear problem (High school simplex method)
- Boolean satisfiability
- Knapsack/travelling salesman/bin packing problem

Reasoning NP

NP: Non-deterministic polynomial time



Scheduling


Forecasting


Planning/Routing


Bin Packing

Complexity class with *very special* properties

- worst case: Hardness scales *exponentially* with size
- most real-world problems: Hardness scales *linearly* with size
- Every problem in NP can be transformed into every other problem in NP
- Boolean satisfiability is lowest level representation and very efficient

Reasoning

Reasoning uses a mathematically exact model

Can prove that solutions exist, or that problem is unsolvable

- Mostly done using Boolean satisfiability (SAT)solvers
  - SAT problems contain (many) Boolean constraints
    - Each a simple assertion: `assert((a and b) or (c and not(b)))`
    - all variables are Booleans (T/F), only boolean operators allowed (`and, or, xor, not`)
    - can encode any integer math problem
- Proofs can be checked in polynomial time (very efficient)
- Proofs can be certified by mathematically rigorous theorem provers
  - Absolute proof! By a machine, a mathematician's dream.

Reasoning

What to wear?

Not naked: pants, shirt, tie
Shirt + tie too formal
Must wear pants + shirt
Can we find satisfying assignment?

A: `assert(p or s or t)`
B: `assert(not(s and t))`
C: `assert(p and s)`
R: `p, s, not(t)`

SAT solvers reason through a problem

- Try out partial assignment, e.g. t=1
- Derive conclusions B: t=1 => s=0, but C: p=s=1, conflict.
  - Check cause of conflict: (t=1), learn opposite: (t=0)
  - Remove from consideration all candidate solutions where (t=1)
    - Prune the decision tree (exponential speedup)

Reasoning

Complete SAT solvers learn new facts

- Absolute, mathematical rigorous, facts
- Valid within the context of the input model
- Learnings are redundant constraints (*derived from input*)
  - Redundant constraints speed up processing
    - Bigger problems => faster processing
    - still NP/NP-Hard, but hardness relative to the size of non-redundant core constraints

Derived facts fill up all available memory (1TB RAM typical)

Reasoning

Real-world ('industrial') problems have
high degree of useful redundance

- = community structure

Hard randomly generated problems do not,
but these problems are of academic interest only.

Reasoning

Three approaches

CDCL: Conflict derived clause learning (Complete)

- reasons + learns new implications
- single thread only!
- very memory hungry

Look-ahead: split large problems into billions of shards (Complete)

- use CDCL and/or local search solver on each shard
- divide and conquer parallelism
- very resource intensive
- speed up limited to hardest shard (maybe as hard as full problem!)

Local search: random(ish) walk through solution space (Incomplete)

- can be parallelised (incl. GPU (my PhD research, 2025))
- cannot prove UNSAT, can only find examples if solvable
- very fast on select SAT problems (LS: seconds, CDCL: years)

Perception vs reasoning



forgot to record the prompt

Perception

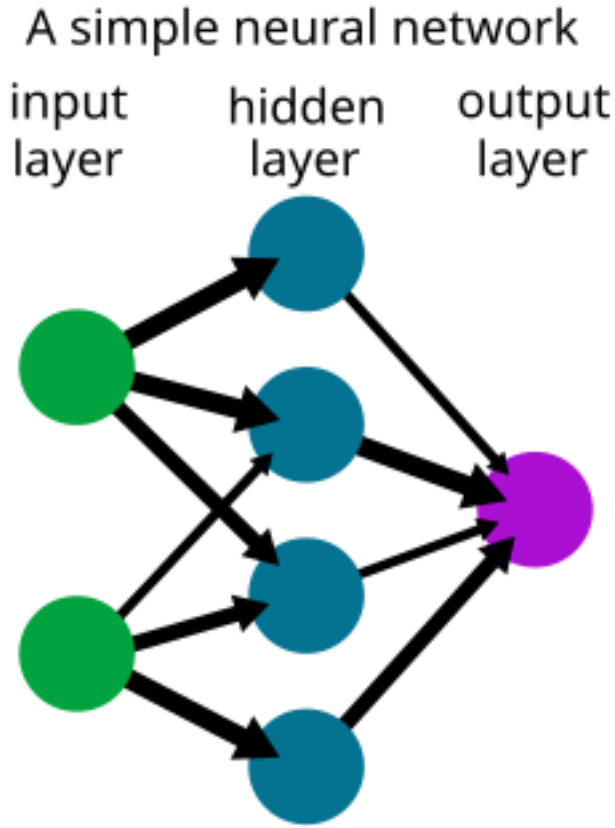
All of generative AI

Build using neural networks

- layers encoded as matrices of floats
- transformations: matrix multiply + add

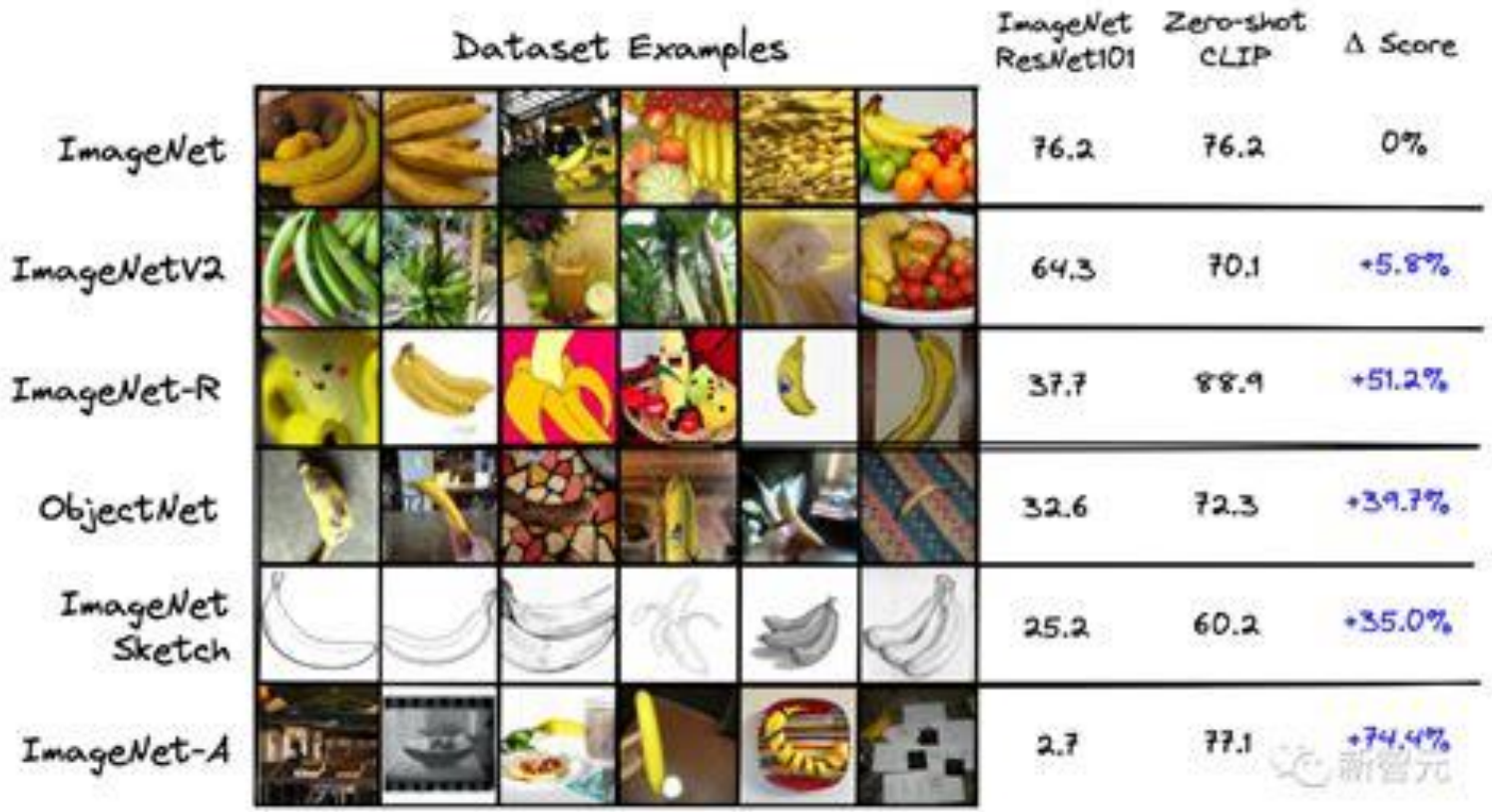
Matrix fused multiply-add: FMULADD

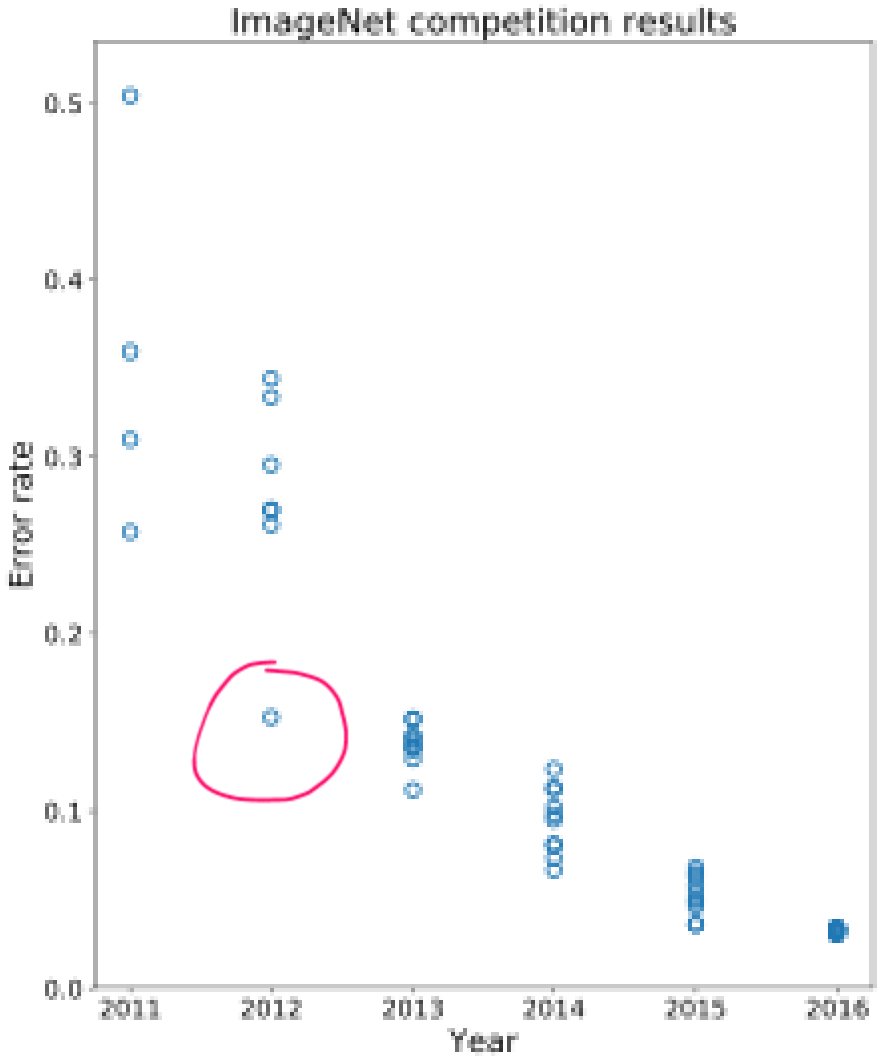
- Matrix size (memory) scales $O(n^2)$
- Workload (operations) scales $O(n^3)$
  - E.g.: for a 100x100 matrix, 100^2 mems, 100^3 ops
    => almost embarrassingly parallel, runs on GPUs



2012 - AlexNet

Watershed moment



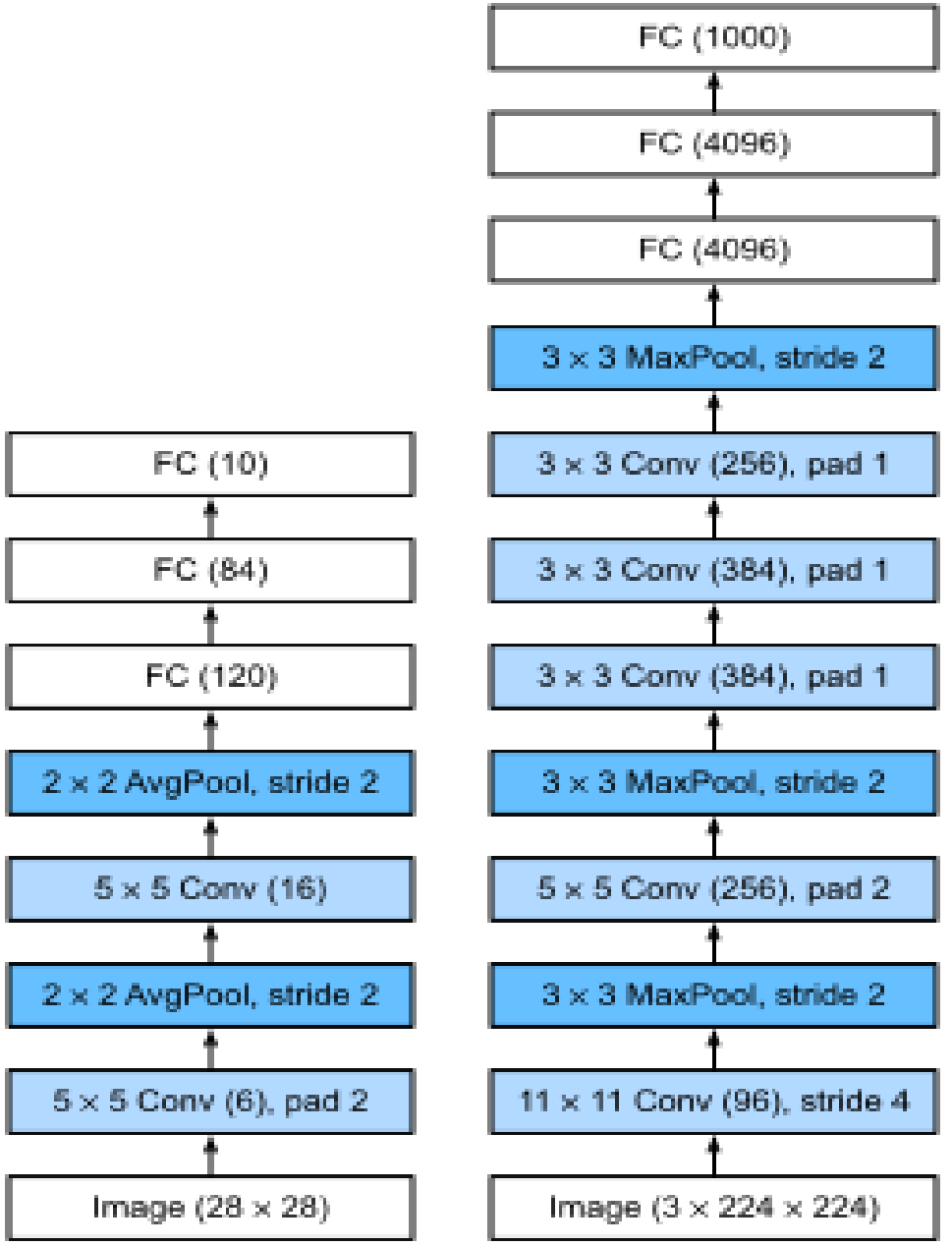


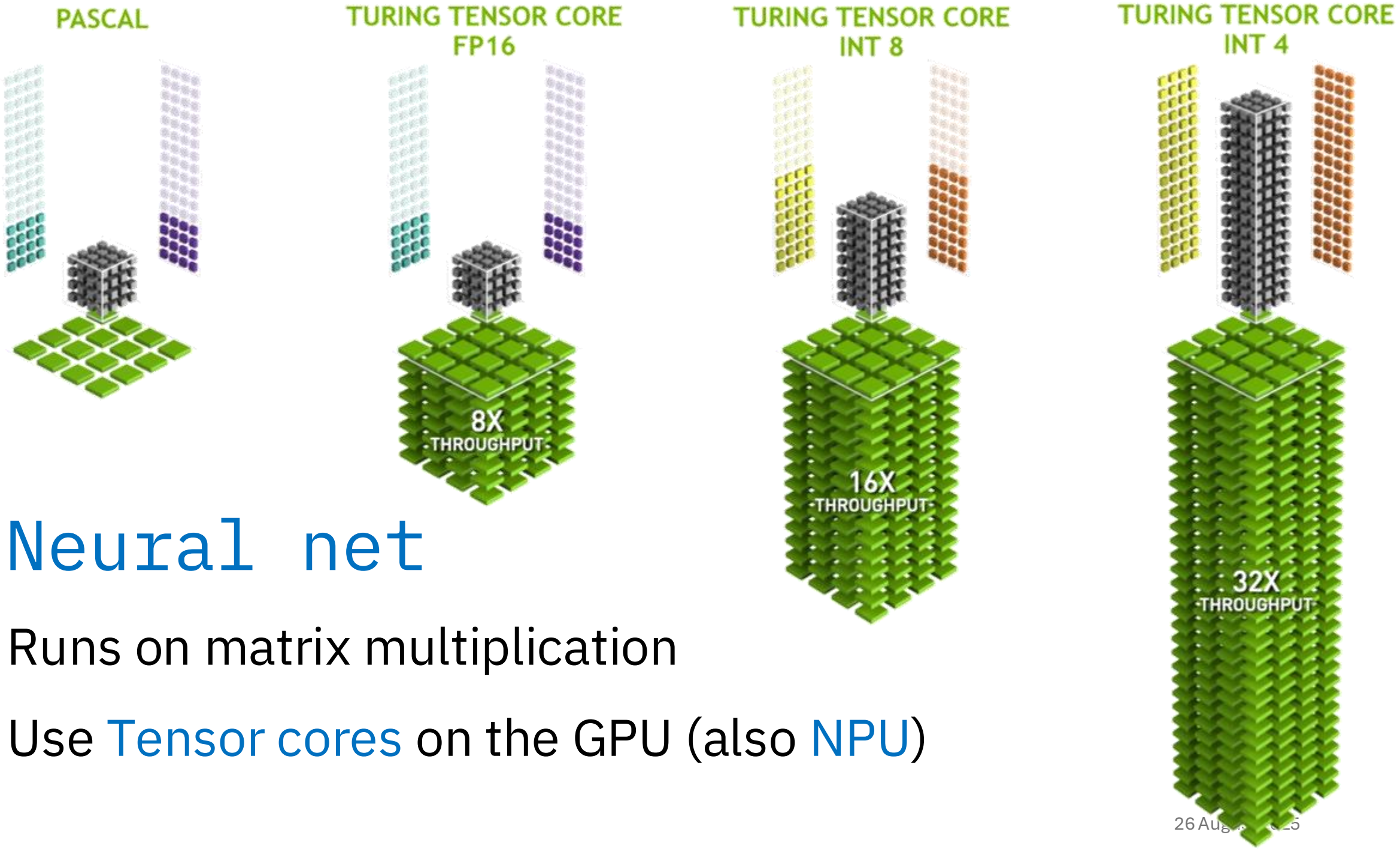
AlexNet

Runs on GPU

- Massively parallel
- 5 Convolutional layers (Conv)
- 3 Fully connected layers (FC)

Much better than shallow neural nets





Neural net

Runs on matrix multiplication

Use Tensor cores on the GPU (also NPU)

Progress

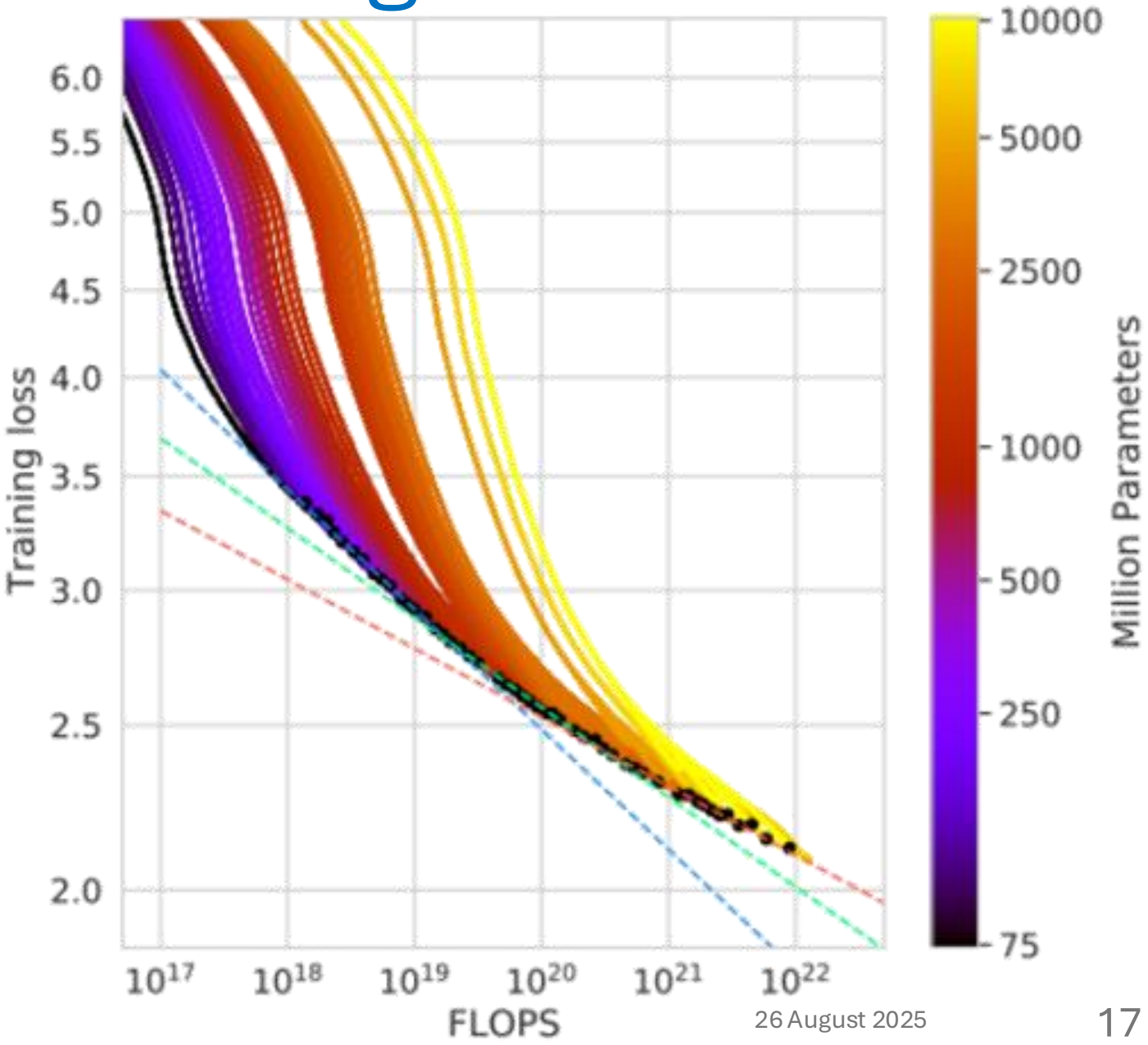
Speed ups in TPUs achieved by

- Bigger TPU/NPUs
- Moar GPUs, bigger models
- Smaller floating-point numbers (float32 -> fl16, fl8, fl4)
  - Leads to loss of precision
  - NVidia claims: int4 just as good as fl32
    - Hard to believe only 16 values == 4 billion options
    - Move to smaller units necessitates new hardware: $$$ NVidia

AI scaling

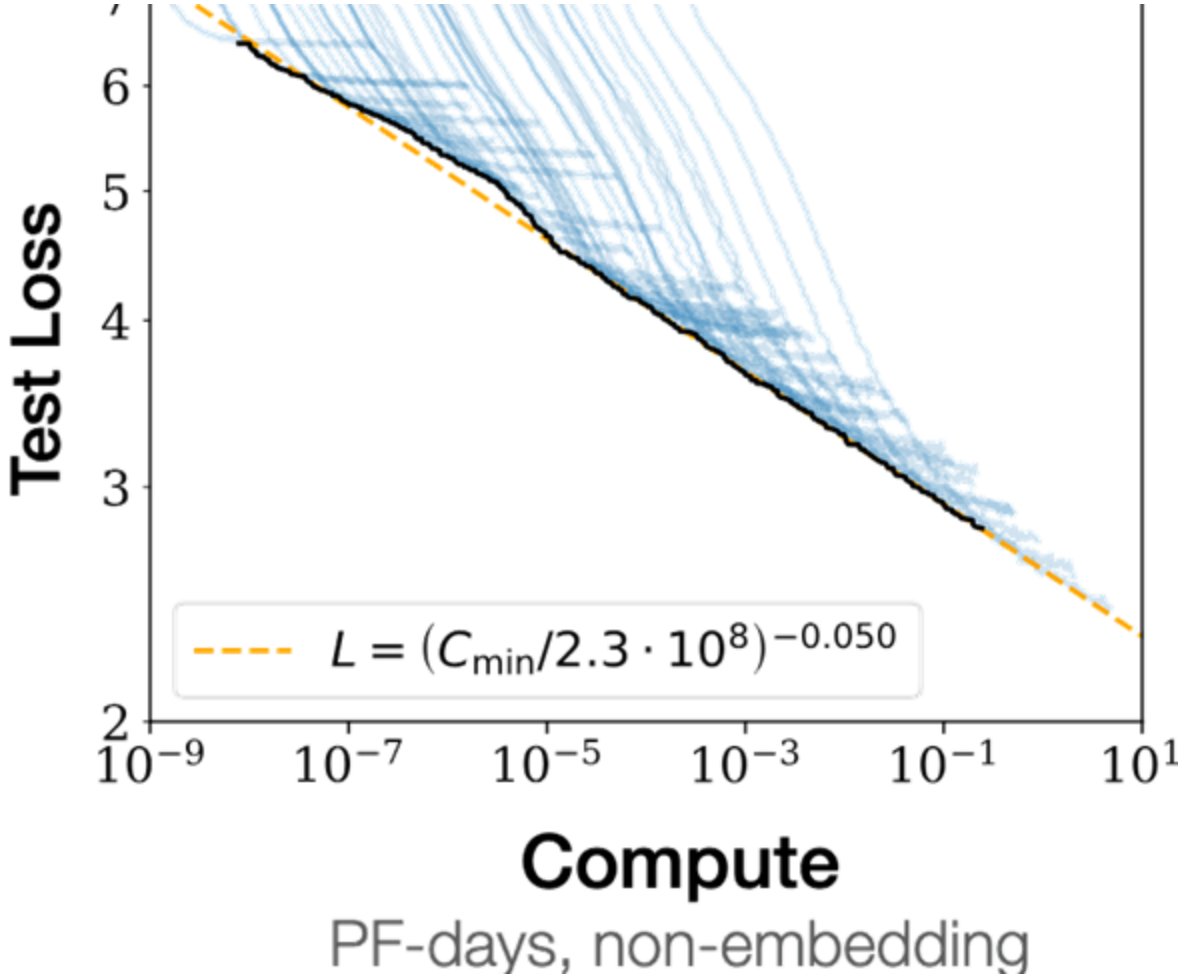
Note the slope of the lines

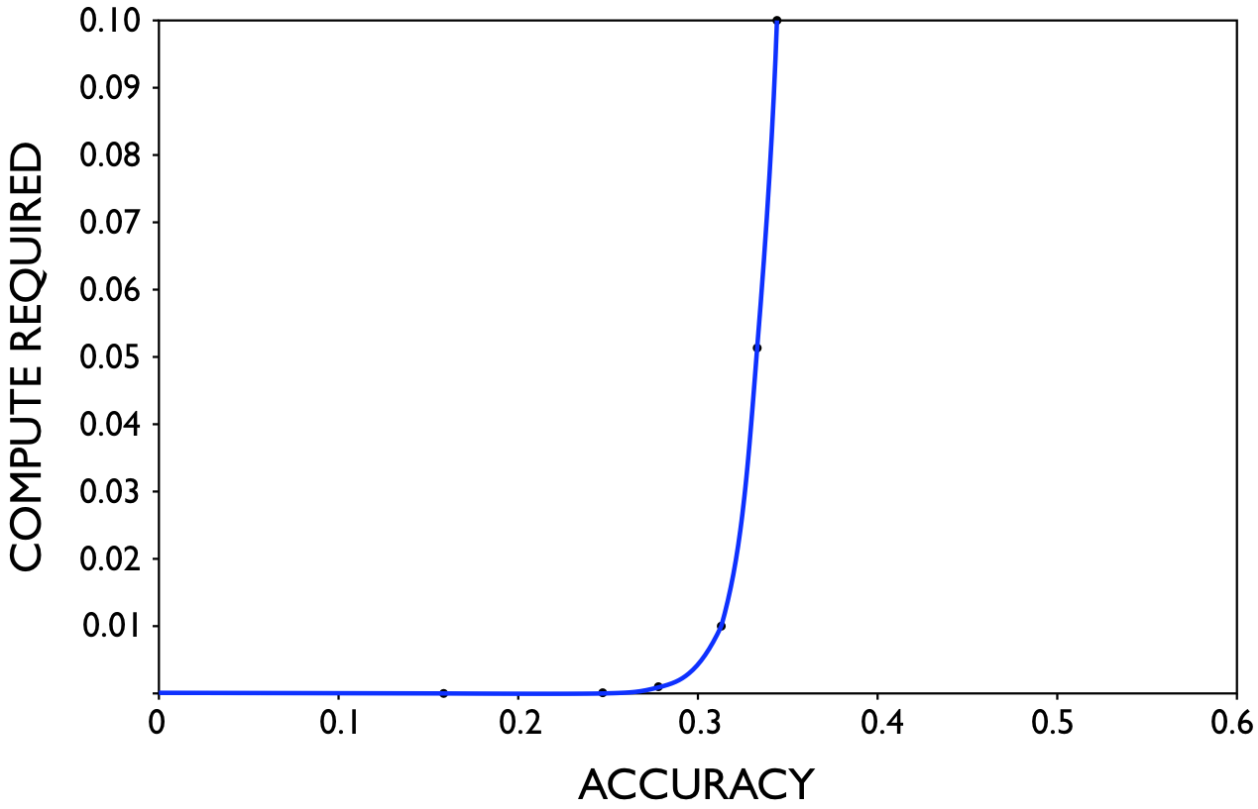
- Log-log plot
- Black line !straight
  - = not power law
- Keeps flattening out
  - diminishing returns
  - terrible scaling



AI scaling diminishing returns

Hype vs reality





terrible returns on investment
minimal improvement for 10x compute

Hype: Scaling Laws for Neural Language Models, OpenAI
Reality: https://www.tobyord.com/writing/the-scaling-paradox

The famous circle r test



Draw a picture containing words and nothing else. There should be five words. Each should stand for a fruit. Circle the letter “r”s.

Images





Abject failure, because picture AI does not understand ‘word’ concept

AI scaling recap

- AI progress as a function of time = impressive
- AI progress as a function of resources is not.

- The scaling laws look smooth and long-lasting
  = proof of poor but predictable scaling
  = !impressive scaling.

- AI quality metrics scale poorly with resources
- the real-world impacts may? scale better.

https://www.tobyord.com/writing/the-scaling-paradox (copy paste mostly, sorry out of time)

AI future?

Current approach scales poorly

- Companies can still throw money at the problem
- 10x scale up of data centers possible (ouch!)

But human brain scales much better

- Perhaps new algorithm/approach can fix issue
- Waiting for new AlexNet discovery

Perception + reasoning?

Humans use perception + reasoning

- No current algorithm known that can combine both
- Perception: knowledge representation is unknown (unknowable?)
- Reasoning: Rigid facts, hard mathematical models
  (Tommie's research works on soft/maybe = defeasible reasoning)
- P: everything fuzzy + R: everything absolute, how to merge?
- Current 'reasoning AI' is just a marketing term
  perception – has no understanding and no way to model concepts

Matrix multiply/AI problems

x * 0 = zero: any zeroed items are gone for good

- Some of this is good, we want to limit scope to Q asked

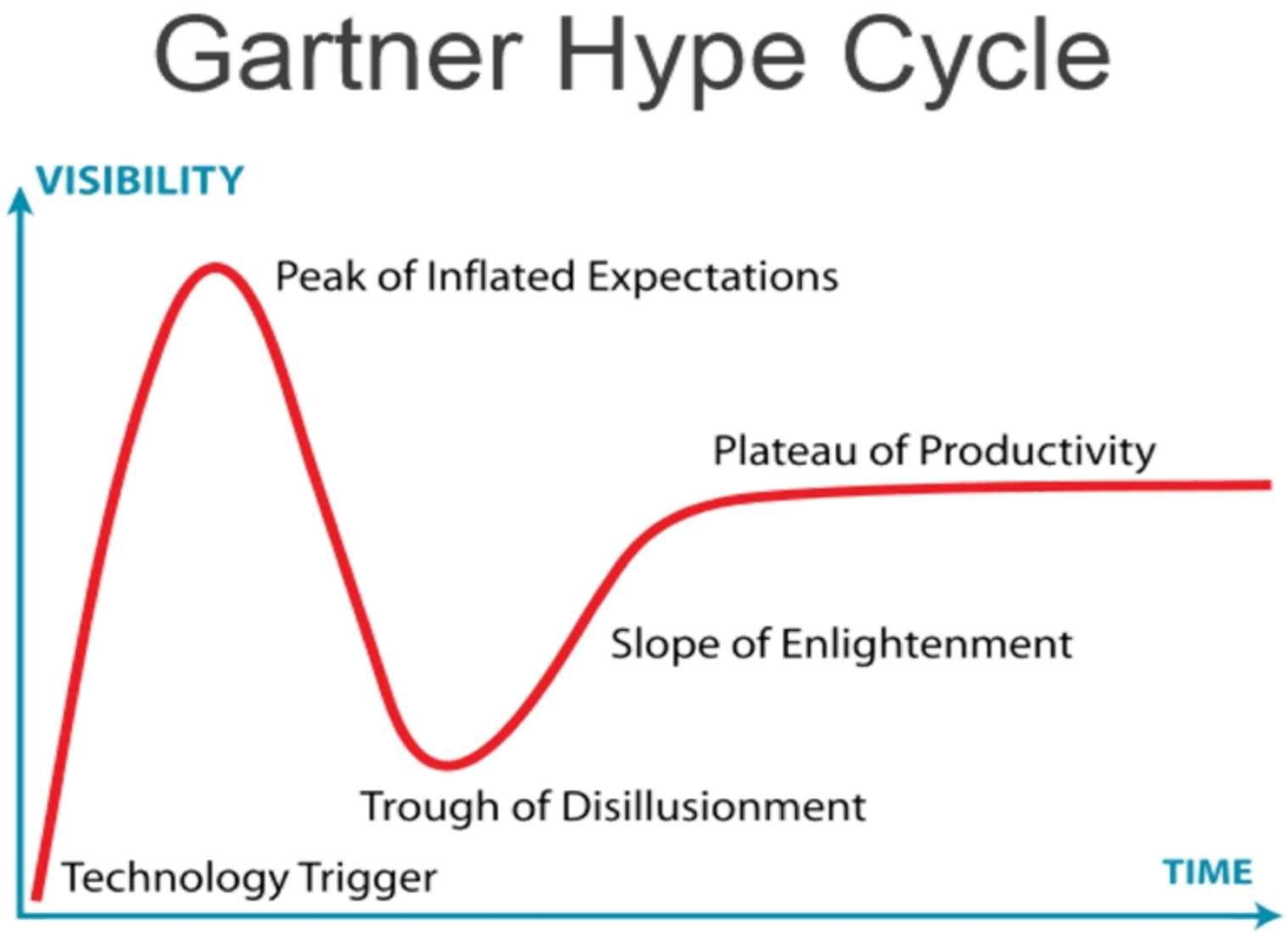
x * inf = inf: inf values cannot be recovered

- I suspect this causes confabulation in AI models

Perception = pattern matching

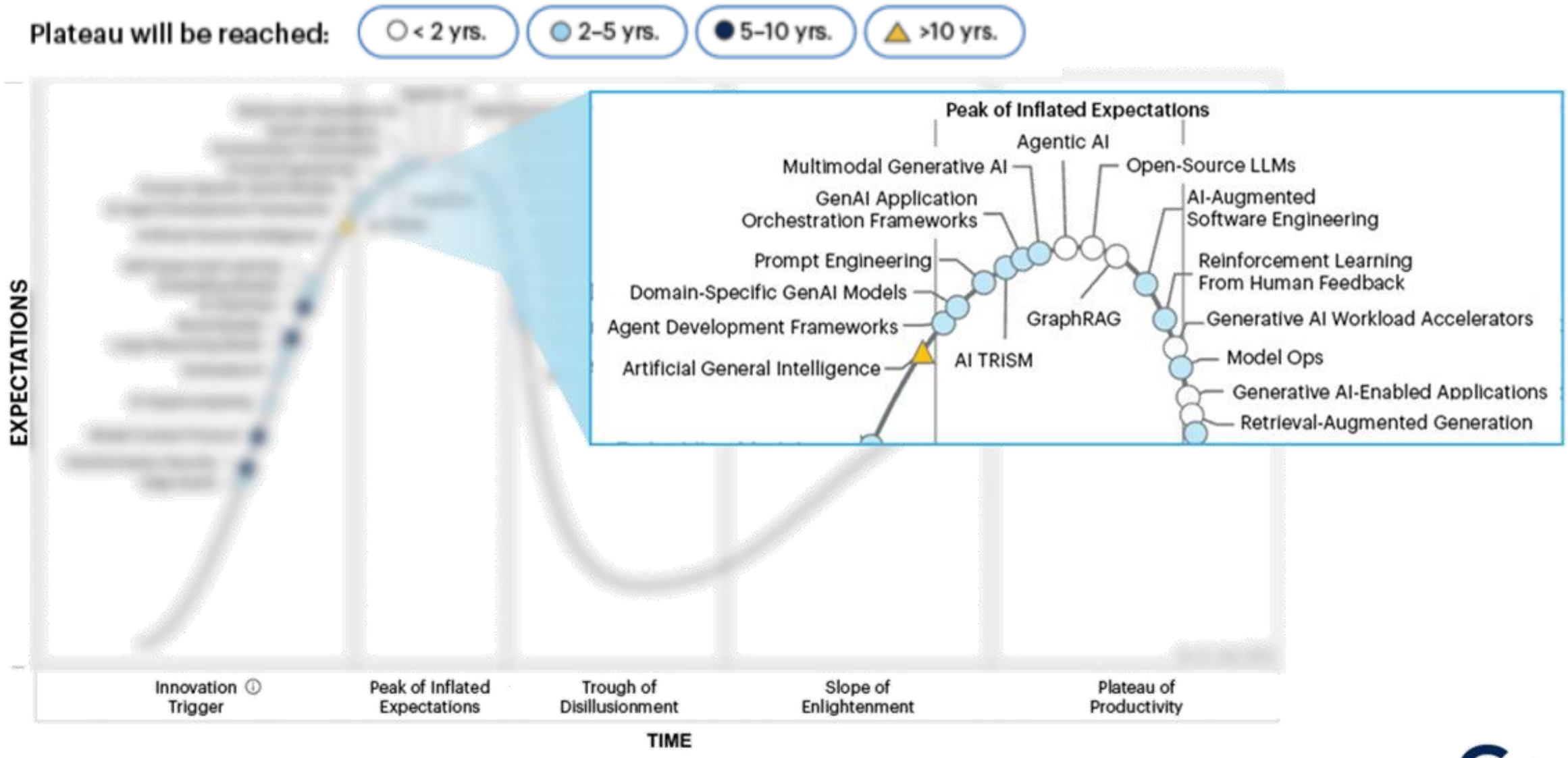
- Extreme patterns get magnified (e.g.: LLM + Nazi-ideology)
- Subtle patterns disappear into noise background (e.g.: LLM + nuance)
- Social networks optimise for engagement (FB: anger good, nuance bad)

New technology: the hype cycle



https://gaswseller.weebly.com/blockchain-gartner-hype-cycle.html

Hype Cycle for Generative AI, 2025



Source: Gartner

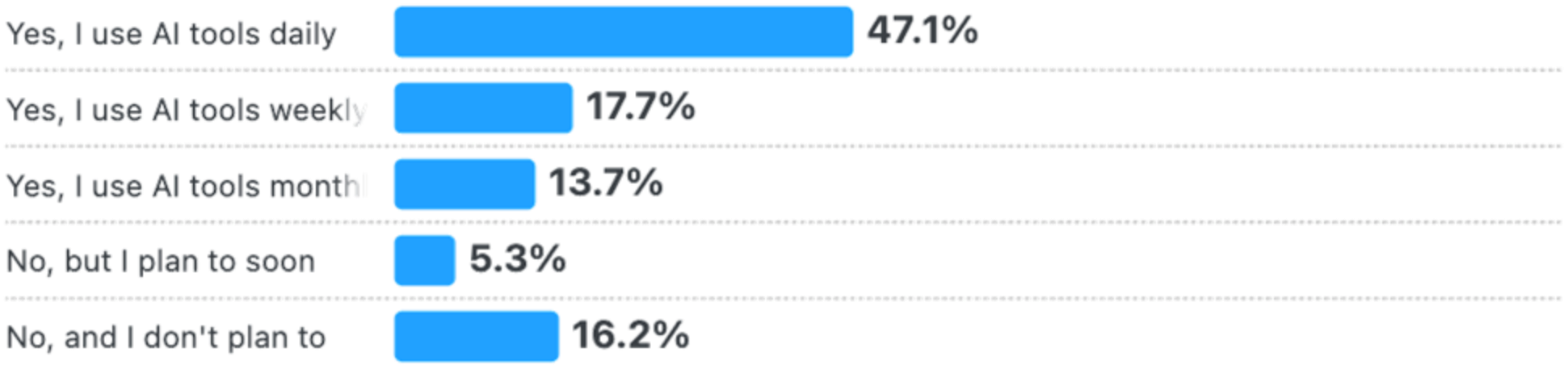
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Gartner

https://emt.gartnerweb.com/ngw/globalassets/en/information-technology/images/hype-cycle-for-generative-ai-2025.png

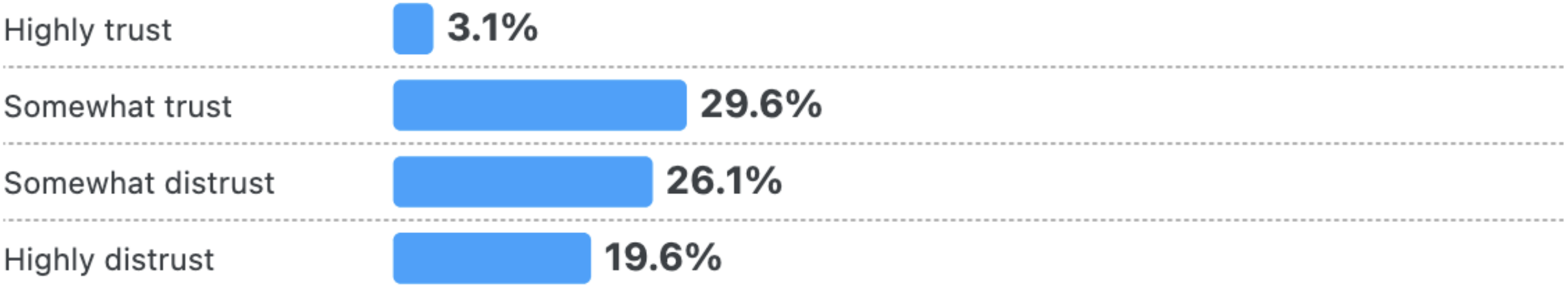
stack overflow developer survey 2025

84% of respondents are using AI tools this year * or planning to use



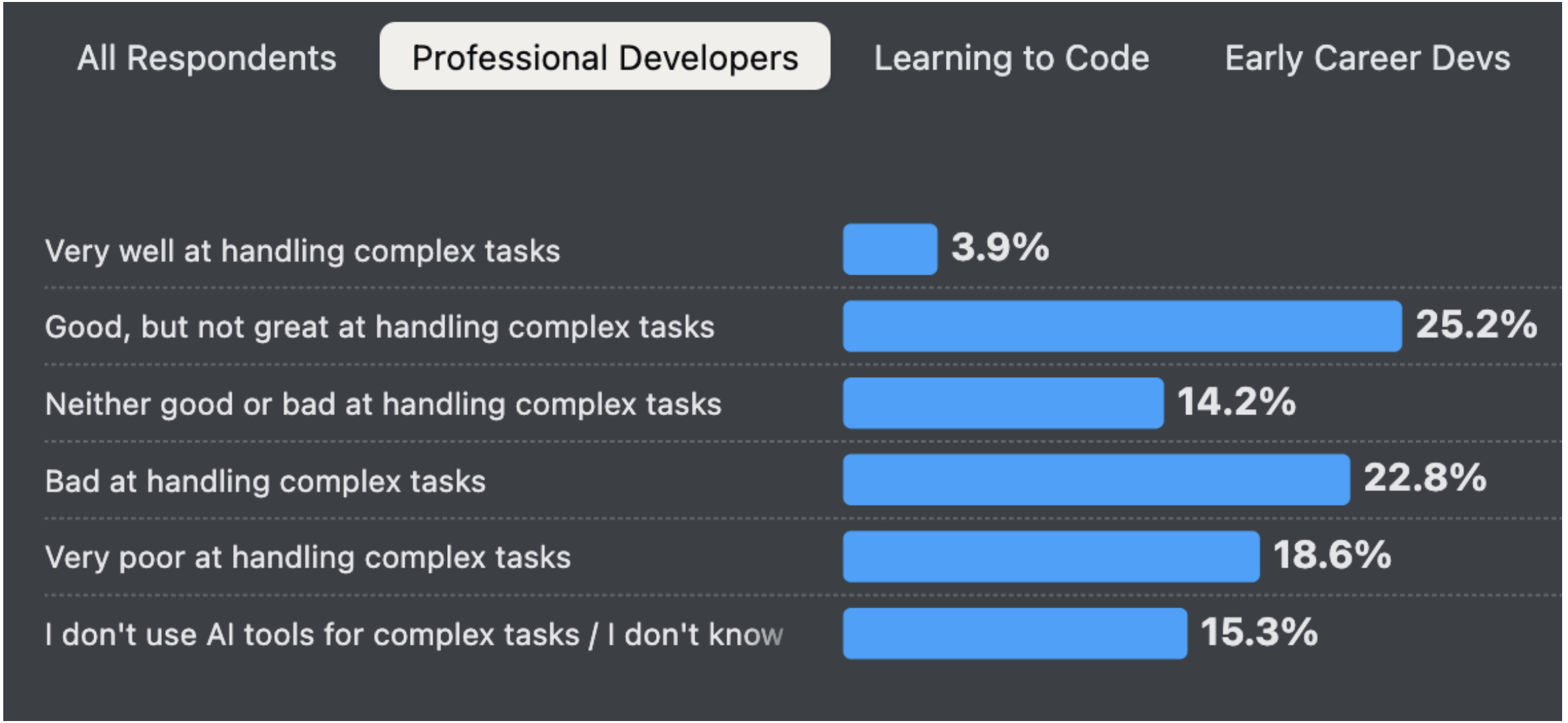
84% of respondents are using or planning to use AI tools in their development process, an increase over last year (76%). This year we can see 51% of professional developers use AI tools daily.

Trust accuracy of AI tools?

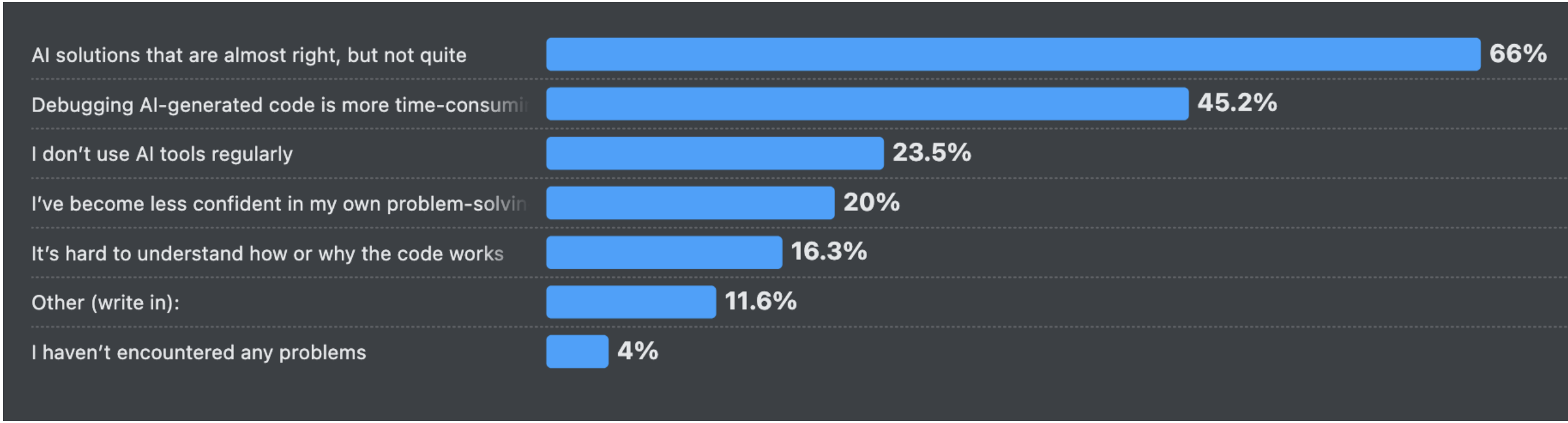


More developers actively distrust the accuracy of AI tools (46%) than trust it (33%), and only a fraction (3%) report "highly trusting" the output. Experienced developers are the most cautious, with the lowest "highly trust" rate (2.6%) and the highest "highly distrust" rate (20%), indicating a widespread need for human verification for those in roles with accountability.

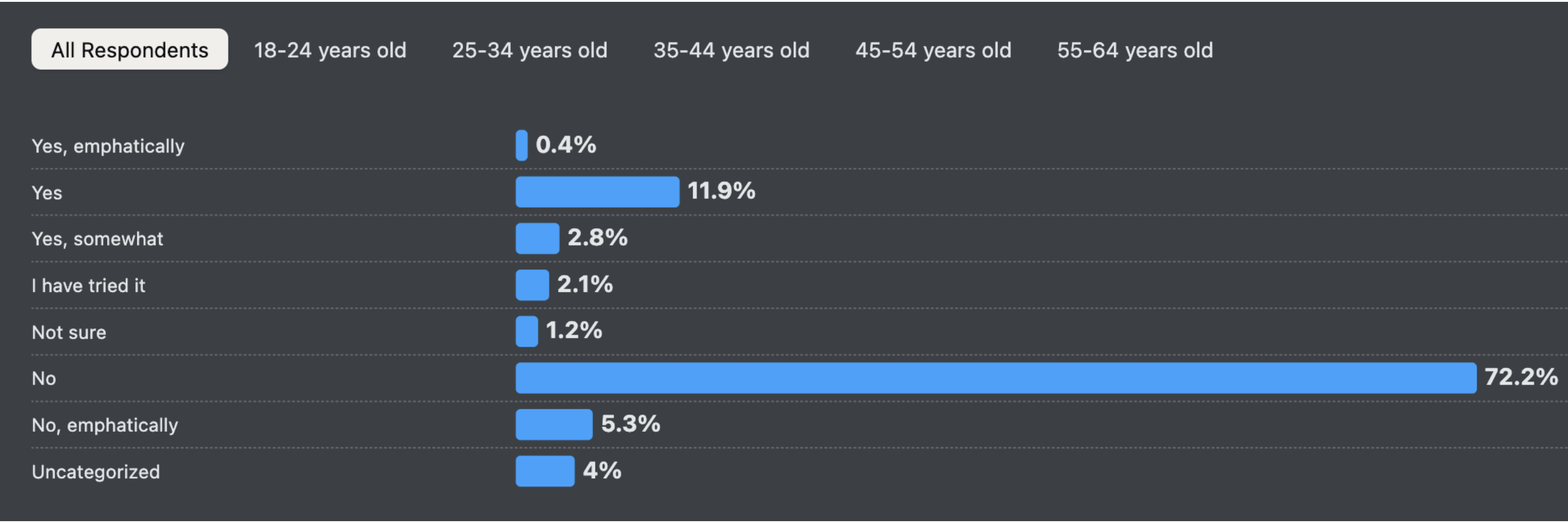
AI ability to handle complex tasks



AI tool frustrations



Vibe coding?


All Respondents
18-24 years old
25-34 years old
35-44 years old
45-54 years old
55-64 years old
Yes, emphatically
0.4%
Yes
11.9%
Yes, somewhat
2.8%
I have tried it
2.1%
Not sure
1.2%
No
72.2%
No, emphatically
5.3%
Uncategorized
4%

UCT CS Generative AI policy...

- Gen AI not declared = plagiarism
  = zero for course + mark on degree
- Vibe coding assignment = no human effort
  = 1% mark for course
- First year CS no Gen AI allowed
- 2 + 3 year: allowed, but must declare + mostly human work
- If explicitly allowed by supervisor: allowed to expand scope of assignment (= more interesting prototypes)
- Science Dean: 80% of mark must come from examination in controlled environment (no internet)

...UCT CS Generative AI policy

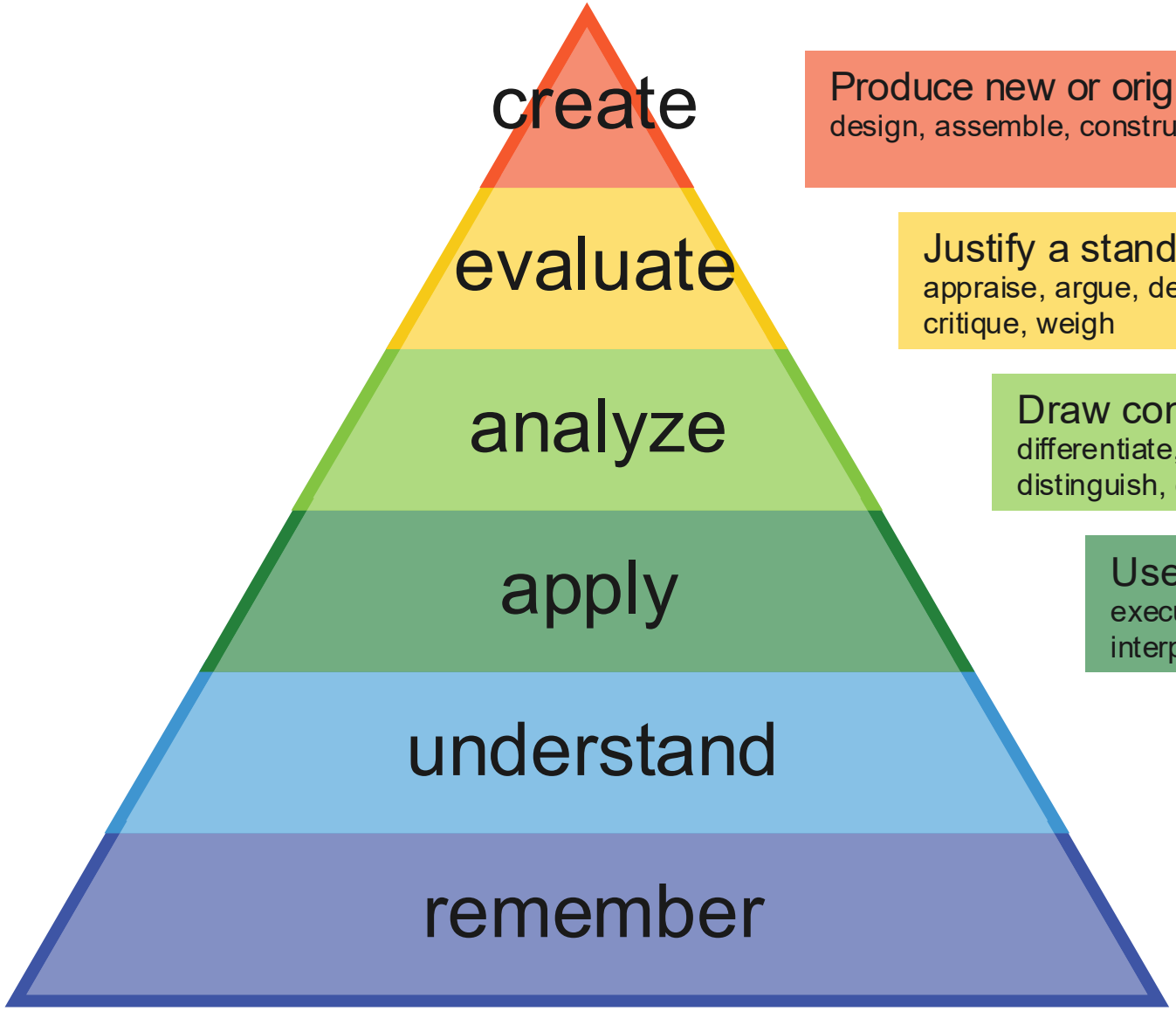
- If student demonstrates insufficient understanding or knowledge of work that he or she claims to have performed themselves, then the grading may be lowered.

Effect of AI on student learning

Research is inconclusive

- inconsistent results
  - supports creation of ideas
  - aids interaction
  - hinders critical thinking and analytical skills

Learning goals: Bloom's taxonomy



Produce new or original work
design, assemble, construct, develop, formulate, investigate, write, present

Justify a stand or decision
appraise, argue, defend, judge, select, support, value, critique, weigh

Draw connections among ideas
differentiate, organize, relate, compare, contrast, distinguish, examine, experiment, question, test

Use information in new situations
execute, implement, solve, use, demonstrate, interpret, operate, schedule, sketch

Explain ideas or concepts
classify, describe, discuss, explain, identify, locate, recognize, report, select, translate

Recall facts and basic concepts
define, duplicate, list, memorize, repeat, state

Bloom's 2021 taxonomy: wikipedia

Meta-study (recap)

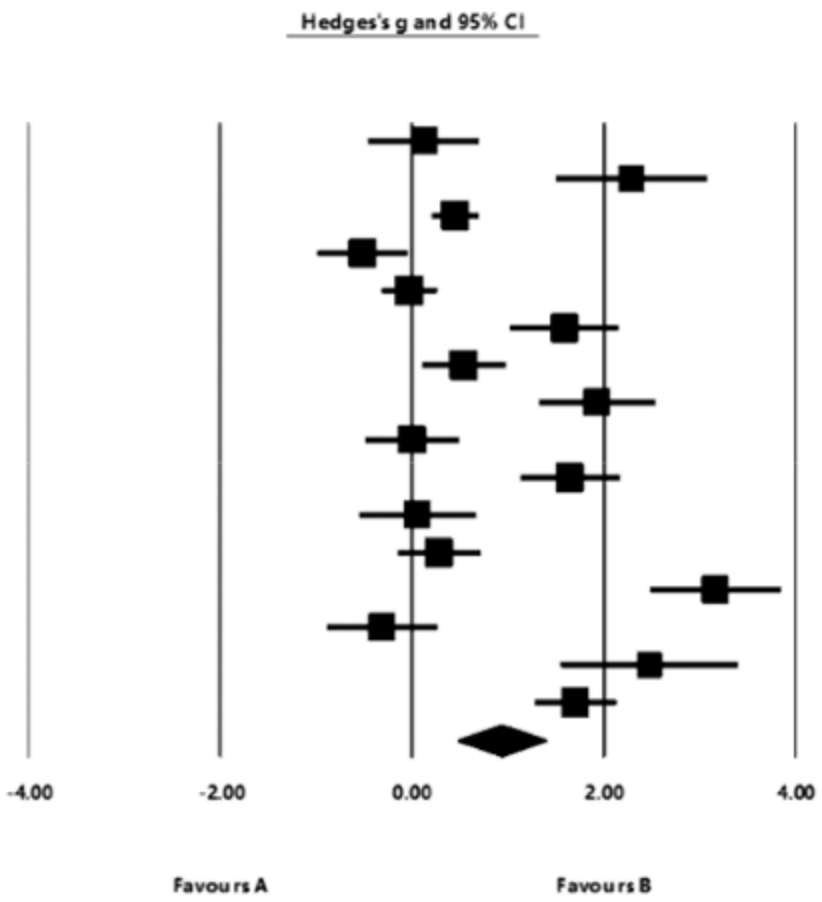
Gen AI 24 studies: mostly positive, but…

- Studies performed in controlled environments
  - Does this translate to messy real-world Uni's?
- Studies with non-significant or negative results less likely to be published peer-reviewed journals
  - also: possible funding bias (big tech)
- Only short-term effects measured
  - GenAI = social experiment, will take decades to play out

Gen AI aids lower order learning?

- A: hinders it
- B: helps it

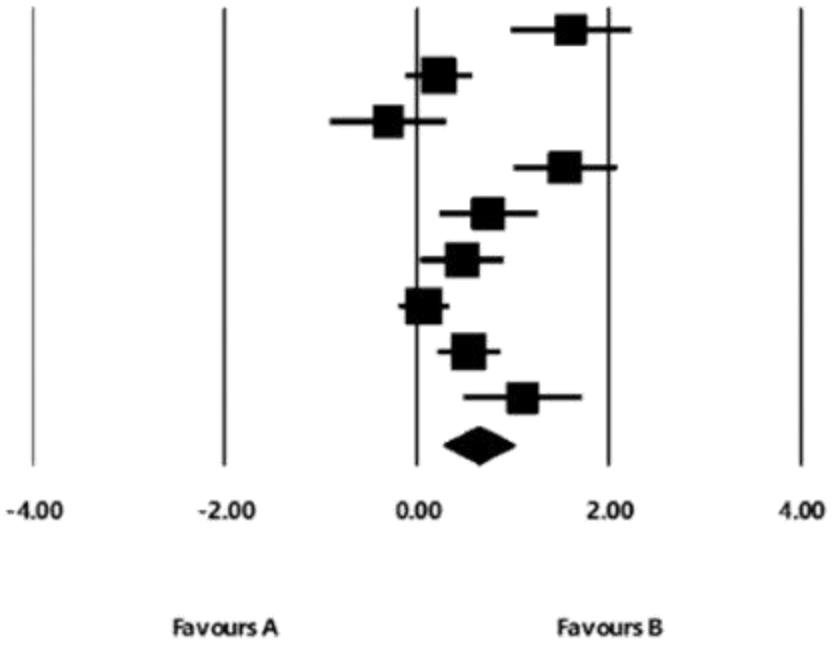
Study name	Outcome	Statistics for each study						
		Hedges's g	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Guo et al.(2023)	combined	0.125	0.296	0.088	-0.456	0.706	0.420	0.674
Hsu et al.(2023)	combined	2.287	0.402	0.161	1.500	3.075	5.692	0.000
Huesca et al.(2024)	basic programming learning gain	0.453	0.127	0.016	0.203	0.702	3.555	0.000
Kavadella et al.(2024)	unannounced blind knowledge examination	-0.514	0.242	0.059	-0.988	-0.040	-2.126	0.034
Kosar et al.(2024)	programming achievements	-0.029	0.148	0.022	-0.318	0.260	-0.196	0.844
Lee et al. (2024)	combined	1.590	0.289	0.083	1.024	2.155	5.508	0.000
Li(2023)	project performance	0.538	0.224	0.050	0.098	0.977	2.398	0.016
Yiliz(2023)	achievement	1.928	0.310	0.096	1.321	2.535	6.225	0.000
Li et al.(2024)	writing performance	0.000	0.253	0.064	-0.496	0.496	0.000	1.000
Boudouaia et al.(2024)	combined	1.648	0.264	0.069	1.131	2.165	6.253	0.000
Mennalla, T. and Quadros(2024)	achivement performance	0.055	0.311	0.097	-0.555	0.665	0.178	0.859
Sun(2024)	programming achievements	0.280	0.220	0.048	-0.151	0.712	1.274	0.203
Zhou and Kim (2024)	combined	3.160	0.347	0.120	2.480	3.841	9.106	0.000
Escalante et al.(2023)	writing test	-0.311	0.297	0.088	-0.893	0.271	-1.046	0.295
Song and Song(2023)	combined	2.472	0.471	0.222	1.549	3.395	5.249	0.000
Urban(2024)	combined	1.702	0.220	0.048	1.271	2.132	7.746	0.000
		0.926	0.234	0.055	0.467	1.384	3.956	0.000



GenAI aids higher order leaning?

- A: hinders it
- B: helps it

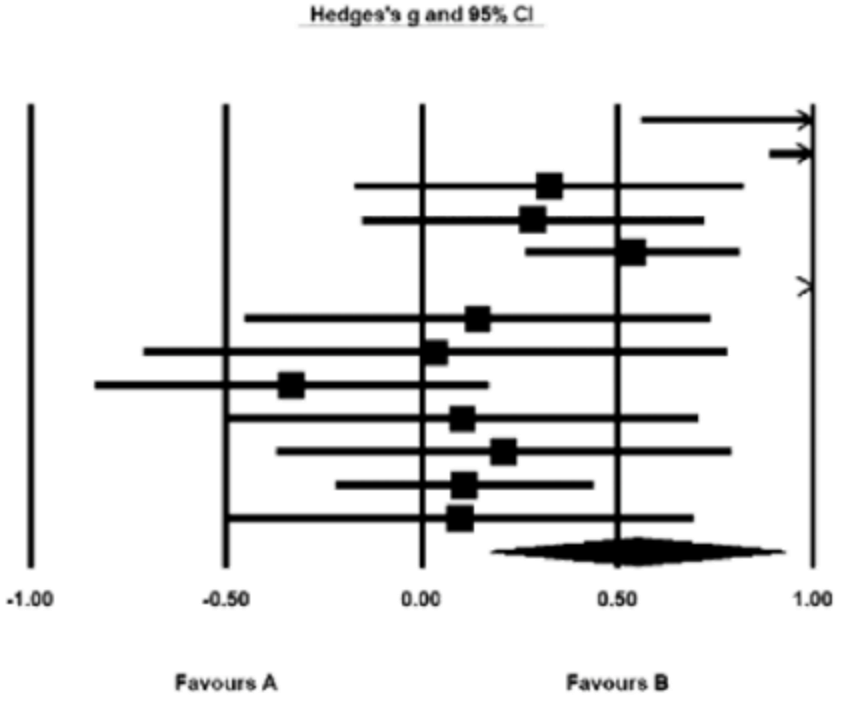
Study name	Outcome	Statistics for each study						
		Hedges's g	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Chang et al.(2024)	combined	1.607	0.321	0.103	0.977	2.237	4.999	0.000
Essel(2024)	combined	0.226	0.179	0.032	-0.124	0.576	1.265	0.206
Guo et al.(2023)	combined	-0.298	0.310	0.096	-0.906	0.310	-0.960	0.337
Hu(2023)	combined	1.543	0.276	0.076	1.002	2.084	5.594	0.000
Lee et al.(2024)	combined	0.748	0.260	0.067	0.239	1.257	2.882	0.004
Li(2023)	Creative thinking tendency	0.473	0.223	0.050	0.035	0.911	2.119	0.034
Lu(2024)	combined	0.077	0.136	0.018	-0.190	0.343	0.563	0.573
Urban(2024)	originality	0.544	0.169	0.028	0.213	0.874	3.224	0.001
Yilmaz and Karaoglan(2023)	computational thinking skills	1.104	0.316	0.100	0.485	1.723	3.496	0.000
		0.640	0.185	0.034	0.277	1.002	3.459	0.001



GenAI aids non-cognitive skills?

- A: hinders it
- B: helps it

Study name	Outcome	Statistics for each study						
		Hedges's g	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Chang et al.(2024)	learning enjoyment	1.153	0.301	0.091	0.562	1.744	3.826	0.000
Hu(2023)	combined	1.420	0.271	0.073	0.889	1.951	5.241	0.000
Lee et al. (2024)	combined	0.327	0.253	0.064	-0.167	0.822	1.297	0.195
Li(2023)	self-efficacy	0.285	0.221	0.049	-0.149	0.719	1.287	0.198
Lu(2024)	self-efficacy	0.537	0.138	0.019	0.266	0.808	3.882	0.000
Ododo et al.(2024)	computer programming self-efficacy	4.167	0.528	0.279	3.132	5.203	7.888	0.000
Woo(2023)	learning motivation	0.144	0.303	0.092	-0.450	0.738	0.475	0.635
Farah et al.(2023)	combined	0.035	0.380	0.144	-0.710	0.780	0.092	0.927
Yiliz(2023)	learning motivation	-0.332	0.257	0.066	-0.835	0.171	-1.294	0.196
Guo et al.(2023)	combined	0.104	0.307	0.094	-0.498	0.706	0.339	0.734
Song and Song(2023)	writing motivation	0.210	0.295	0.087	-0.368	0.788	0.711	0.477
Urban(2024)	combined	0.110	0.167	0.028	-0.218	0.437	0.655	0.512
Yilmaz and Karaoglan(2023)	combined	0.098	0.304	0.092	-0.497	0.693	0.324	0.746
		0.552	0.192	0.037	0.176	0.928	2.879	0.004



AI code security

45% of gen AI code
= known insecure!

Better models?
Syntax has ↑,
Security not.

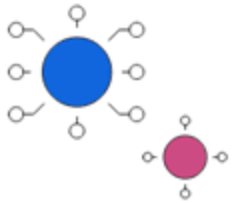
The highlights of our findings are as follows:



Across all models and all tasks, **only 55% of generation tasks result in secure code**. In other words, in 45% of the tasks the model introduces a known security flaw into the code.

Security performance has been **largely unchanged over time**, even as models get better at generating syntactically correct code.





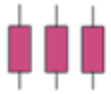
Larger models **do not perform significantly better** than smaller models



Security performance **varies dramatically** by CWE type.

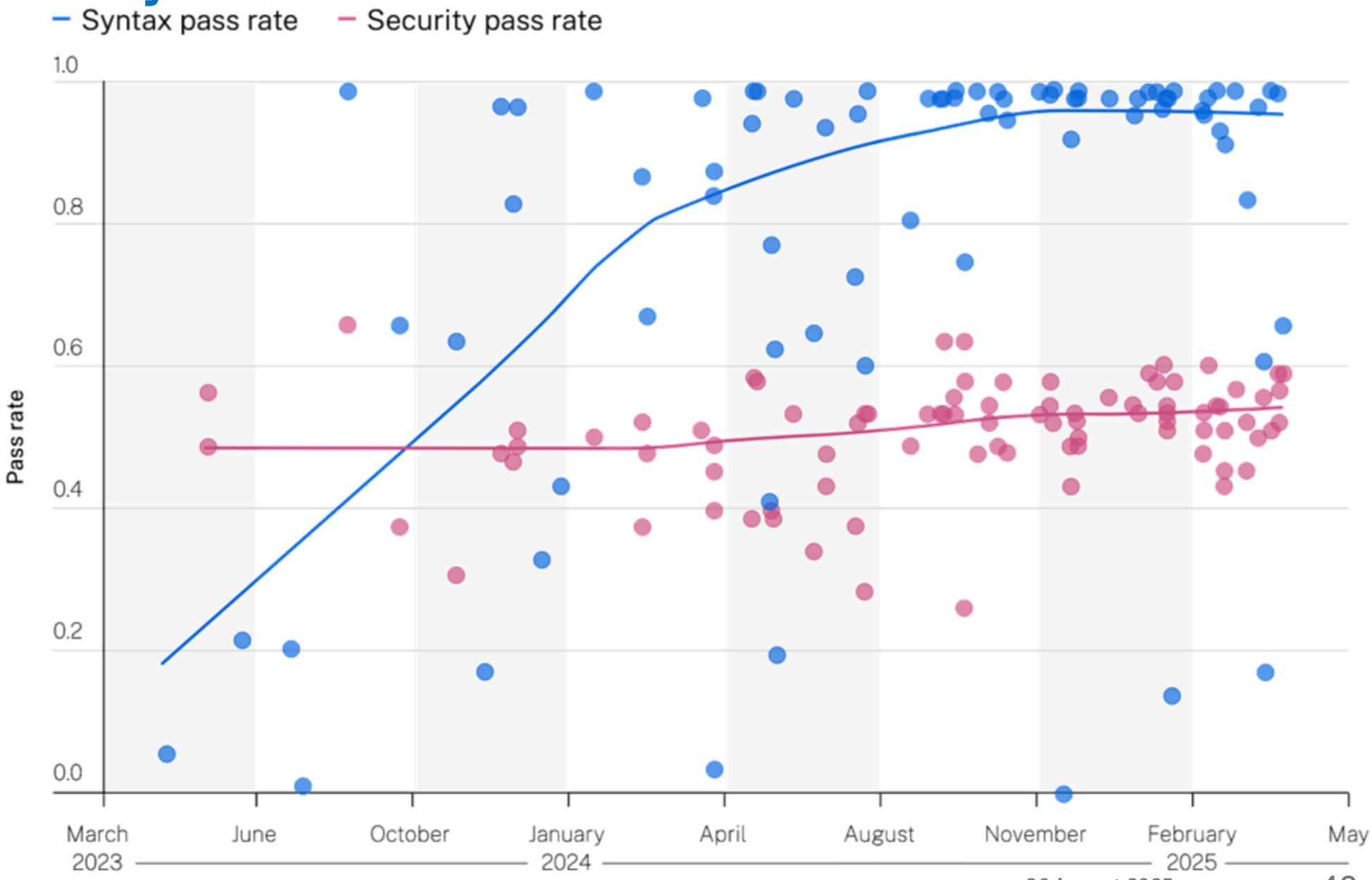


Security performance **varies somewhat** across languages.



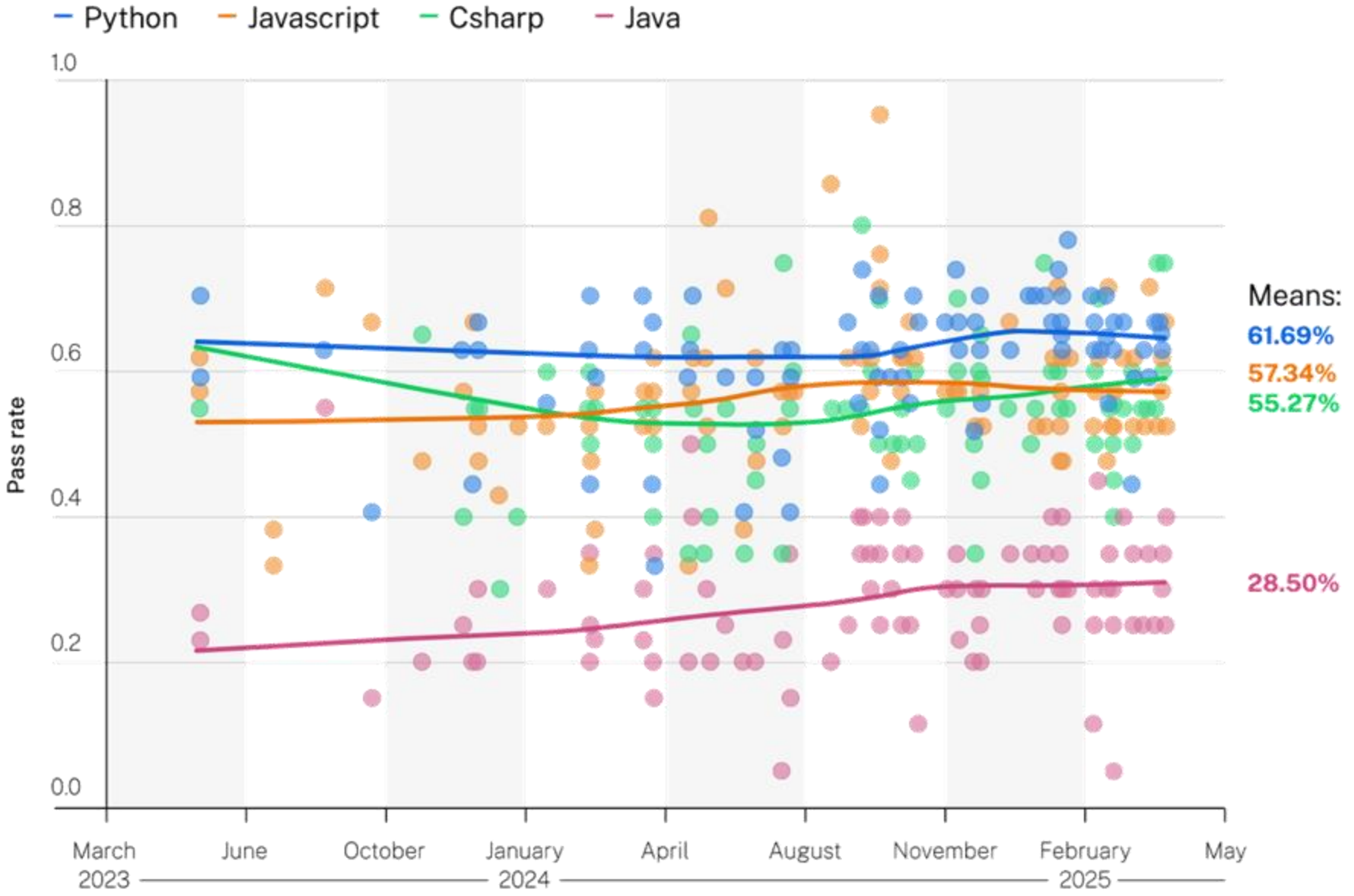
Security performance is **remarkably consistent** across models.

Gen AI Security



https://www.veracode.com/wp-content/uploads/2025_GenAI_Code_Security_Report_Final.pdf

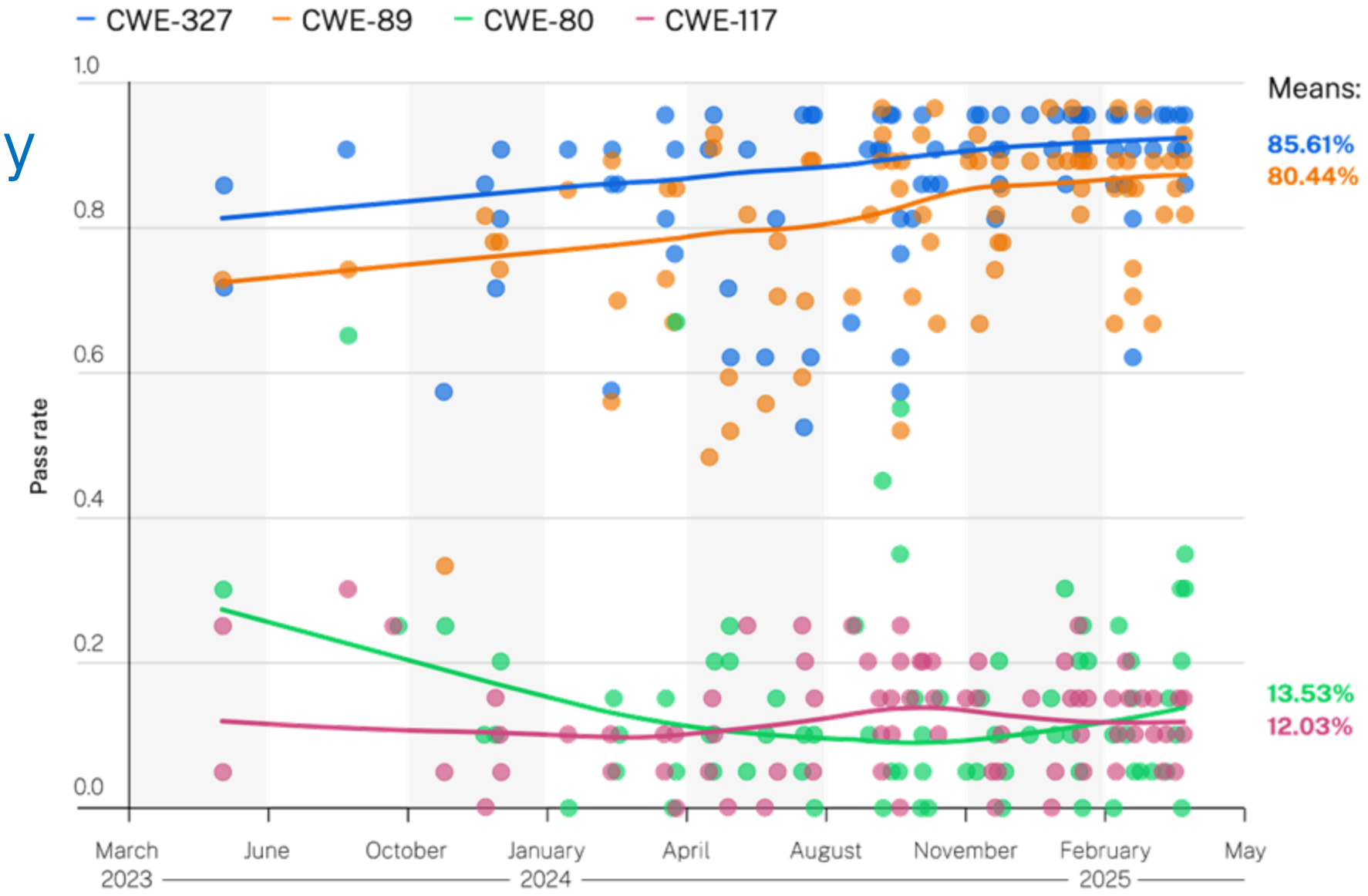
Secure pass-rate per language



https://www.veracode.com/wp-content/uploads/2025_GenAI_Code_Security_Report_Final.pdf

Pass rate over 4 issues

- CWE-327
  Bad cryptography
- CWE-89
  SQL injection
- CWE-80
  XSS
- CWE-117
  Log injection



GenAI = insecure code, bugs?

- Proxy for logical bugs in code?
  - Likely: GenAI lacks understanding
  - Interesting research:
    - GenAI code vs bugs found by sanitizer
- Use of GenAI shifts effort
  - from writing functionality => finding/fixing bugs
    - writing code = 5% effort
    - finding fixing bugs = 50% effort
  - does not look like progress